UAV-Enabled Asynchronous Federated Learning

Zhiyuan Zhai, Xiaojun Yuan, *Senior Member, IEEE*, Xin Wang, *Fellow, IEEE*, and Huiyuan Yang, *Graduate Student Member, IEEE*

***Abstract*— To exploit unprecedented data generation in mobile edge networks, federated learning (FL) has emerged as a promising alternative to the conventional centralized machine learning (ML). By collectively training a unified learning model on edge devices, FL bypasses the need of direct data transmission, thereby addressing problems such as latency issues and privacy concerns inherent in centralized ML. However, in practical deployment FL suffers from low learning efficiency due to the involved straggler issue and huge uplink overhead. In this paper, we develop a UAV-enabled over-the-air asynchronous FL (UAV-AFL) framework to address this problem. This framework significantly enhance the learning efficiency by supporting the UAV as the parameter server (UAV-PS) in collecting data over-the-air and updating model continuously. We conduct a convergence analysis to quantitatively capture the impact of model asynchrony, device selection and communication errors on the UAV-AFL learning efficiency. Based on this analysis, a unified communication-learning problem is formulated to maximize asymptotical learning accuracy by optimizing the UAV-PS trajectory, device selection and over-the-air transceiver design. Simulation results reveal valuable insights for the system design and demonstrate that the proposed UAV-AFL scheme achieves substantially improvement in learning efficiency compared with the state-of-the-art approaches.**

***Index Terms*— Asynchronous federated learning, UAV communication, over-the-air computation, staleness, device selection, trajectory optimization.**

I. Introduction

EXTENSIVE data available on mobile edge devices have sparked considerable interest in the development of artificial intelligence (AI) services, such as image recognition and natural language processing [1], [2]. However, traditional machine learning (ML) requires the collection of data in a centralized server for model training, which incurs substantial energy and bandwidth costs, significant time delays, and potential privacy concerns [3]. Federated learning (FL), as a novel ML paradigm, has emerged to tackle these problems [4]. In a typical FL framework, each edge device computes local model updates based on its individual dataset and subsequently transmits the model update to a parameter server (PS) for model aggregation. The global model is then updated at the PS and shared with the participating devices. In this way, FL replaces direct data transmission with the model updates (gradients) transmission, which not only alleviates the communication burden but also safeguards the confidentiality of local data.

Despite the above advantages, the learning efficiency of FL is very low in some practical deployments, e.g., remote areas or military operations [5], [6]. The huge uplink communication overhead incurred in the learning process significantly impairs the efficiency of FL [4]. Over-the-air computation [7], [8] for FL uplink model aggregation is a promising technique to reduce this communication cost. By using the superposition property of electromagnetic waves in wireless channels, edge devices can concurrently transmit their gradients utilizing the same time-frequency resources. The authors in [7] thus introduce over-the-air model aggregation for FL, and the result demonstrates substantial communication efficiency improvement and latency reduction when compared to orthogonal multiple access (OMA) schemes. This approach is extended to multiple-input multiple-output (MIMO) scenarios in [8] where over-the-air computation is applied.

In addition to the uplink burden, a fundamental challenge, namely the straggler issue [9], [10] makes FL less efficient. Since edge devices have different channel qualities and computing abilities, the straggler issue is generally twofold. Devices with poor channel conditions bring communication errors to model aggregation and devices with inferior computing abilities need longer time to compute the gradient, both jeopardizing the learning efficiency of FL. In this context, using an unmanned aerial vehicle (UAV) as PS (UAV-PS) is a promising approach to tackle both of these problems. By utilizing the high mobility and on-demand deployment ability of UAV [11], [12], [13], [14], UAV-PS can flexibly balance the difference between computing time and ensure reliable communication links with devices suffering poor channel conditions. These advantages substantially help to improve the learning efficiency. Several works have investigated the UAV-assisted FL systems [15], [16], [17], [18]. The authors in [15] propose to deploy a UAV as a network relay, where the UAV collects model updates from devices and then forward them to the model owner for increasing the coverage. In order to minimize a fairness metric regarding device computing time, a UAV-PS trajectory optimization is carried out in [16].

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Zhiyuan Zhai and Xin Wang are with the Department of Communication Science and Engineering, Fudan University, Shanghai 200438, China (e-mail: xwang11@fudan.edu.cn).

Xiaojun Yuan and Huiyuan Yang are with the National Key Laboratory of Wireless Communications, University of Electronic Science and Technology of China, Chengdu 611731, China (e-mail: xjyuan@uestc.edu.cn).

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The UAV-assisted FL system using over-the-air computation has been investigated in [17] and [18]. In particular, [17] introduces a hierarchical aggregation approach where UAV-PS collects gradients from devices at various locations and merges them at the end of each iteration. [18] dispatches the UAV-PS to assist the aggregation of different learning models and determines the transmission scheduling using greedy deflation approach. However, these studies [15], [16], [17], [18], though successfully integrating UAV into the FL system, cannot effectively improve the learning efficiency due to a pivotal issue: The synchronous structure of FL is inherently incompatible with the asynchronous nature of UAV-assisted communication systems. Specifically, in the FL system, the model aggregation is only finished when all the gradients of devices are collected. However, the UAV-PS can only collect the gradients of the nearby devices (i.e., a subset of all devices). This procedural incompatibility leads to a significant delay in model update until the UAV-PS navigates through the entire service area and collects gradients from all devices, which severely impairs the overall efficiency of the learning process. A feasible solution to mitigate this incompatibility is to allow the UAV-PS to perform one FL iteration exclusively with the nearby devices. In this approach, the UAV-PS broadcasts the global model to these adjacent devices, and the latter then use the model for gradient computation. Subsequently, the UAV-PS collects the computed gradients and updates the model parameters. After that, the UAV-PS moves to another area and begins new FL iteration with the nearby devices. However, while this scheme reduces the latency of model updates, it requires the UAV-PS to hover over the serviced devices when they conduct local gradient computation. This time-consuming local computation process[1] could also adversely impact the overall learning efficiency.

To fully unleash the system potential and maximize the learning efficiency, we develop a novel UAV-enabled over-the-air asynchronous FL (AFL) scheme. This scheme is generally obtained by further changing the order of the model aggregation and broadcast steps in the framework of aforementioned solution. Since the aggregation happens before the broadcast step, some aggregated gradients are computed based on different versions of the global model, which leads to an AFL problem. Specifically, in the proposed UAV-enabled over-the-air AFL (UAV-AFL) scheme, the UAV-PS selects some devices to communicate with, collects these devices' gradients over-the-air, updates the model and then performs model broadcasting to these selected devices. After that, these selected devices conduct local computation based on the newly received model while the UAV-PS continues the next round of AFL by selecting new devices without waiting for the local computations of these previously selected devices. Compared with conventional UAV-assisted FL settings [15], [16], [17], [18], the UAV-AFL scheme improves learning efficiency by offering a compatible asynchronous communication-learning design which allows the UAV-PS to traverse the service area to collect gradients and update models continuously.

In this paper, we propose the UAV-AFL framework to improve the FL learning efficiency. We develop a unified analysis framework that captures the impact of model asynchrony, device selection and communication errors on the AFL training loss. Based on this, a joint communication-learning optimization algorithm is formulated to maximize the asymptotic learning performance. Our main contributions are summarized as follows.

- We study a UAV-assisted FL system where a UAV-PS is deployed to perform over-the-air aggregation over line-of-sight (LoS) channels. We propose a novel UAV-AFL scheme to improve the learning efficiency. This scheme has two advantages: 1) The UAV-PS can immediately updates the global model upon the reception of the aggregated gradients. 2) The UAV-PS can collect the newly computed gradients for model update when devices are conducting local computation.
- We introduce a metric called 'staleness upper bound', i.e., the maximal version discrepancy among the models maintained by all edge devices, to characterize the level of asynchrony in the UAV-AFL system. Then, a rigorous convergence analysis on the learning loss is carried out by taking the model asynchrony, device selection and communication errors into account. To the best of our knowledge, this is the first attempt to analyze the convergence of asynchronous learning/optimization in the presence of communication error and device selection loss.
- We formulate a communication-learning co-design optimization problem to enhance the learning efficiency of the UAV-AFL based on the convergence analysis. An effective algorithm is developed to solve the problem. Specifically, we transform the original problem via geometric programming (GP), and design a two-layer penalty-based algorithm using successive convex approximation (SCA) principle to jointly optimize the UAV trajectory, device selection and transceiver design.

Simulation results under different system settings provide an in-depth study of the trade-offs between over-the-air aggregation quality, model update rates and asynchronous level, which offers valuable insights for the UAV-AFL system design. The comparison results indicate that the proposed UAV-AFL scheme achieves a substantial improvement in learning efficiency compared to the existing state-of-the-art solutions.

Notations: We denote the real and complex number sets by $\mathbb{R}$ and $\mathbb{C}$, respectively. We use $(\cdot)^{\mathrm{T}}$ to denote the transpose; $|\mathcal{S}|$ to denote the cardinality of set $\mathcal{S}$; $\circ$ to denote the hadamard product; $\mathbb{E}$ to denote the expectation operator; $\|\cdot\|$ to denote the l_2-norm; $\mathcal{CN}(\mu, \sigma^2)$ to denote circularly-symmetric complex normal distribution with mean μ and covariance σ^2.

II. System Model

A. Federated Learning System

We consider a FL system with M devices collaboratively training a common global model, which can be characterized

[1]In the internet of things (IoT) networks, edge devices often require several seconds for completing a single gradient descent (GD) algorithm due to their limited CPU capacity [19], [20], which is significantly greater than the gradient transmission time.

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TABLE I
LIST OF SYMBOLS

Symbol	Description
M	Number of devices
D	Dimension of the model parameters
$\boldsymbol{x}$	Model parameter
$\mathcal{S}$	Entire training dataset
$\mathcal{S}_m$	Local training dataset of device m
$F(\mathbf{x})$	Global empirical loss function
$f_m(\mathbf{x})$	Local loss function of device m
$\mathbf{g}_m(k)$	Gradient computed by device m at slot k
$\tau_{m,k}$	Staleness of device m at slot k
δ_t	Duration of each slot
$\mathbf{q}(k)$	Coordinates of the UAV-PS at slot k
$v_{\max}$	Maximum flight speed of the UAV-PS
$a_{\max}$	Maximum acceleration of the UAV-PS
$\mathbf{q}_F$	Initial and final position of the UAV-PS
$h_m(k)$	Uploading channel of device m at slot k
g_0	Channel power gain at distance $d_{\text{ref}} = 1$m
$d_m(k)$	Distance of UAV-PS to device m at slot k
$\vartheta_m(k)$	Phase shift of $h_m(k)$
$\mathcal{N}_k$	Devices that have LoS links at slot k
λ	Learning rate
P_0	Maximum transmit power of devices
σ_n^2	Noise power
L	Lipschitz constant
μ	Strong convexity parameter
δ	Bounded variance parameter
α_1, α_2	Parameters for bounding the gradient

by a D-dimensional model parameter $\mathbf{x} \in \mathbb{R}^D$. The entire training data set $\mathcal{S}$, with size $|\mathcal{S}|$, is distributed over the edge devices. We use $\mathcal{M} = \{1, \cdots, M\}$ to denote the set of devices involved in the system. Suppose that each device possesses a local training data set $\mathcal{S}_m$ with size $|\mathcal{S}_m| = S_d$, where $\sum_{m\in\mathcal{M}} |\mathcal{S}_m| = MS_d = |\mathcal{S}|$ and $\mathcal{S}_i \cap \mathcal{S}_j = \varnothing, \forall i \neq j$. The learning objective of the FL system is to find an optimal parameter $\mathbf{x}^\star$ minimizing the empirical loss function given by

$$F(\mathbf{x}) = \frac{1}{|\mathcal{S}|} \sum_{\xi_i \in \mathcal{S}} f(\mathbf{x}, \xi_i), \tag{1}$$

where ξ_i is the i-th training sample including the input feature and output label, and $f(\mathbf{x}, \xi_i)$ is the loss function with respect to model $\mathbf{x}$ based on sample ξ_i. Denote the local loss function of device m as

$$f_m(\mathbf{x}) = \frac{1}{|\mathcal{S}_m|} \sum_{\xi_i \in \mathcal{S}_m} f(\mathbf{x}, \xi_i). \tag{2}$$

Then, the FL task in (1) can be represented as

$$\min_{\mathbf{x}\in\mathbb{R}^d} \; F(\mathbf{x}) = \frac{1}{M} \sum_{m\in\mathcal{M}} f_m(\mathbf{x}). \tag{3}$$

B. UAV-Assisted Communication System

A UAV-assisted communication network is used to fulfill the FL task, where a UAV-operated parameter server (i.e., UAV-PS) is deployed to coordinate the edge devices performing collaborative training. We employ a three-dimensional Cartesian coordinates system, where the coordinates of the m-th ground device are $\mathbf{w}_m = [x_m, y_m, 0], \forall m$. We assume that the UAV-PS maintains a constant flight altitude H and the position of the UAV-PS remains constant during one time slot with duration δ_t. The coordinates of the UAV-PS at time slot k, denoted by $\mathbf{q}(k)$, can be expressed as $\mathbf{q}(k) = [x(k), y(k), H]$. We then have the following mechanical constraints

$$\mathbf{v}(k) = \frac{\mathbf{q}(k+1) - \mathbf{q}(k)}{\delta_t}, \quad \|\mathbf{v}(k)\| \le v_{\max}, \; \forall k, \tag{4}$$

$$\mathbf{a}_k = \frac{\mathbf{v}(k+1) - \mathbf{v}(k)}{\delta_t}, \quad \|\mathbf{a}_k\| \le a_{\max}, \; \forall k, \tag{5}$$

$$\mathbf{q}(0) = \mathbf{q}(K) = \mathbf{q}_F, \tag{6}$$

where $v_{\max}$ (or $a_{\max}$) denotes the maximal speed (or acceleration) of the UAV-PS, K is the total number of time slots, and $\mathbf{q}_F$ denotes the initial and final dispatch position of the UAV-PS. We assume that the channel state information (CSI) is available at edge devices and the UAV-PS [21], and the Doppler effect arising from the movement of the UAV-PS can be effectively compensated at the receiver [22]. Moreover, due to the ground-to-air nature of UAV communication, we focus on the line-of-sight (LoS) channels for the connection links. Therefore, the channel links connecting the devices to the UAV-PS adhere to the free-space path loss model, expressed as

$$h_m(k) = \sqrt{g_0 d_m^{-2}(k)}\vartheta_m(k), \quad m \in \mathcal{N}_k, \; \forall k \tag{7}$$

where $h_m(k)$ is the channel between the m-th device and the UAV-PS in the k-th time slot, g_0 represents the channel power gain at the reference distance $d_{\text{ref}} = 1$m, $d_m(k) = \|\mathbf{q}(k) - \mathbf{w}_m\|$ denotes the distance between device m and the UAV-PS at the k-th time slot, $\vartheta_m(k) = e^{j\theta_m(k)}$ with $\theta_m(k)$ being the phase shift of $h_m(k)$, and $\mathcal{N}_k$ denotes the set of devices that have LoS communication links with the UAV-PS during time slot k. Note that $\mathcal{N}_k$ can be determined based on practical channel conditions.

III. UAV-Enabled Over-the-Air AFL Framework

A. Motivation

In the conventional UAV-assisted FL literature [15], [16], [17], [18], the system conducts the following steps iteratively: 1) The UAV-PS broadcasts the global model to all devices. 2) All devices conduct local gradient computation based on the received model. 3) The UAV-PS navigates through the entire service area and collects gradients from all devices. 4) The UAV-PS finally aggregates the gradients and use it to update the global model. Here, the UAV-PS can only update the global model after all devices have completed local computations and the UAV-PS has finalized the gradient collection of all devices (i.e., after step 3 and 4). Hence, this approach needs excessively long time to carry out a single model update, which significantly impairs learning efficiency.

To address this issue, a straightforward choice is to allow each FL iteration to involve only the devices located near the UAV-PS. In this method, the UAV-PS traverses the entire

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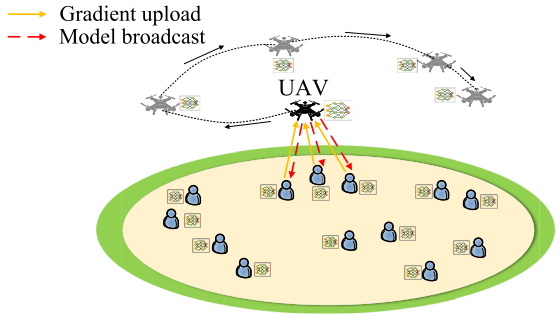


Fig. 1. UAV-enabled AFL system model.

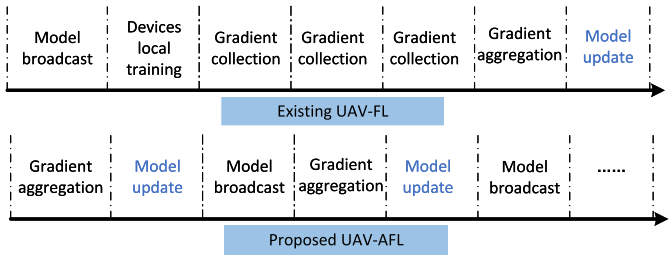


Fig. 2. The learning process of UAV-AFL and traditional UAV-FL.

service area and provides FL service sequentially to the nearby devices. When the UAV-PS approaches some devices, the system conducts the following steps: 1) The UAV-PS broadcasts the global model to the near devices. 2) These devices compute the gradients locally. 3) The UAV-PS collects the newly computed gradients from these devices. 4) The UAV-PS aggregates the gradients and updates the global model. In this way, the UAV-PS can directly utilize the gradients from near devices to update the global model, thereby enhancing the learning efficiency. However, to collect the gradients (step 4), the UAV-PS in this approach cannot do anything but hover near the devices to wait for them completing local computations, which also compromises the system efficiency.

Therefore, to improve the overall learning efficiency, it is crucial to develop a strategy where the UAV-PS can operate without interruption. This strategy should enable the UAV-PS to collect gradients and update models continuously for ensuring a seamless and efficient learning process.

B. Proposed UAV-Enabled AFL System

We consider a UAV-enabled AFL system as depicted in Fig. 1. In this system, at each time slot, the UAV-PS selects a subset of devices to communicate with. Specifically, the UAV-PS collects the gradients of the selected devices, updates the global model, and then broadcast the updated model to these devices. Subsequently, the selected devices initiate computation based on the received model, while the UAV-PS continues to collect gradients from other devices for updating the model at the same time. Denote by $\mathcal{J}_k \subset \mathcal{M}$ the set of aggregation-ready devices in the k-th time slot, i.e., the devices that complete their local computation and are ready for aggregation at time slot k. In the k-th time slot, the system conducts the following procedures:

- *Device selection*: The UAV-PS selects a subset of devices in $\mathcal{J}_k \cap \mathcal{N}_k$ to participate in the aggregation process. Let $\mathcal{M}_k$ be the set of selected devices in the k-th time slot. We define $a_m(k)$ as an indicator function of the selection status of the m-th device in the k-th time slot. That is, $a_m(k) = 1$ if $m \in \mathcal{M}_k$, and $a_m(k) = 0$ otherwise.
- *Gradient aggregation*: The devices in $\mathcal{M}_k$ transmit the computed gradients $\mathbf{g}_m(k), m \in \mathcal{M}_k$ to the UAV-PS via the wireless channels. Here, $\mathbf{g}_m(k)$ is represented as

$$\mathbf{g}_m(k) = \nabla f_m(\mathbf{x}(k - \tau_{m,k})), \quad m \in \mathcal{M}_k, \tag{8}$$

where $\mathbf{x}(k - \tau_{m,k})$ denotes the global model of time slot $k - \tau_{m,k}$, $\nabla f_m(\mathbf{x}(k - \tau_{m,k}))$ is the gradient of device m with respect to the model $\mathbf{x} = \mathbf{x}(k - \tau_{m,k})$, and $\tau_{m,k} = k - \max_{k'<k}\{k' \mid m \in \mathcal{M}_{k'}\}$ is referred to as *staleness* [23] of device m in time slot k. Guided by the received signals, the UAV-PS obtains an estimate $\hat{\mathbf{g}}(k)$ of the desired aggregated gradient $\mathbf{g}(k)$. The desired gradient $\mathbf{g}(k)$ is given by

$$\mathbf{g}(k) = \frac{1}{|\mathcal{M}_k|} \sum_{m \in \mathcal{M}_k} \mathbf{g}_m(k). \tag{9}$$

However, due to the impact of channel fading and communication noise, we in general have $\hat{\mathbf{g}}(k) \neq \mathbf{g}(k)$. With $\hat{\mathbf{g}}(k)$, the global model $\mathbf{x}(k)$ is then updated by

$$\mathbf{x}(k+1) = \mathbf{x}(k) - \lambda \hat{\mathbf{g}}(k), \tag{10}$$

where λ is the learning rate.

- *Model broadcast*: The UAV-PS broadcasts the updated global model $\mathbf{x}(k+1)$ to the devices $m \in \mathcal{M}_k$.
- *Local computation*: The devices $m \in \mathcal{M}_k$ compute their individual gradients using $\mathbf{x}(k+1)$. Let c_m represent the computation time of the m-th device. Hence, the devices $m \in \mathcal{M}_k$ cannot communicate with the UAV-PS from time slot $k+1$ to $k + c_m$ inclusive.

Remark 1: In the synchronous FL setting [24], [25], [26], all the gradients $\mathbf{g}$*'s are guaranteed to be computed using the unified global model* $\mathbf{x}(k)$*. However, in the UAV-enabled AFL system, some aggregated gradients* $\mathbf{g}_m(k)$ *might be computed based on the stale global model, i.e., the model* $\mathbf{x}(k')$*, where* $k' < k$*.*

Remark 2: As shown in Fig. 2, compared to the approaches in [15], [16], [17], and [18], the UAV-AFL scheme provides a more efficient learning strategy. Here, the UAV-PS can seamlessly collect the newly computed gradients from edge devices and update the model parameters immediately. This feature not only accelerates the UAV-PS update rate but also increases the device computing efficiency.

C. Over-the-Air Model Aggregation

We employ the over-the-air computation technique [7] to achieve efficient aggregation in the UAV-AFL system. Specifically, in the k-th time slot, the selected devices $m \in \mathcal{M}_k$ collaboratively transmit their gradients $\{\mathbf{g}_m(k) : m \in \mathcal{M}_k\}$ using the same time-frequency resources during this time slot. By manipulating the transmit and receive equalization coefficients, the estimated weighted sum $\hat{\mathbf{g}}(k)$ in (10) can

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be coherently superposed at the UAV-PS. The specifics are elaborated as follows.

To perform over-the-air aggregation, the initial step is the normalization of $\mathbf{g}_m(k), m \in \mathcal{M}_k$, which can be expressed as

$$\tilde{\mathbf{g}}_m(k) = \left(\mathbf{g}_m(k) - \bar{g}_m(k)\mathbf{1}_{D\times 1}\right)/\sqrt{v_m(k)}, \quad (11)$$

where $\tilde{\mathbf{g}}_m(k) \in \mathbb{R}^D$ is the normalized zero-mean and unit-variance version of $\mathbf{g}_m(k)$, and $\bar{g}_m(k) = \frac{1}{D}\sum_{d=1}^{D} g_m^{(k)}(d)$ as well as $v_m(k) = \frac{1}{D}\sum_{d=1}^{D}\left(g_m^{(k)}(d) - \bar{g}_m(k)\right)^2$ are the mean and variance of $\mathbf{g}_m(k)$, respectively. By following the common practice as in [9] and [27], we assume $\{\bar{g}_m(k), v_m(k) | m \in \mathcal{M}_k\}$ to be transmitted to the UAV-PS via error-free links. We then modulate (analog modulate) $\tilde{\mathbf{g}}_m(k)$ as a complex version $\mathbf{r}_m(k) \in \mathbb{C}^C$ for transmission, i.e.,

$$\mathbf{r}_m(k) = \tilde{\mathbf{g}}_m(k)\left(1 : \frac{D}{2}\right) + j\tilde{\mathbf{g}}_m(k)\left(\frac{D+2}{2} : D\right), \quad (12)$$

where we assume D is even and $C = \frac{D}{2}$. Let $\beta_m(k)$ denote the pre-processing factor of device m at time slot k. Then the transmit signal $\mathbf{s}_m(k)$ can be expressed as

$$\mathbf{s}_m(k) = \beta_m(k)\mathbf{r}_m(k) \in \mathbb{C}^C, \quad (13)$$

and the corresponding power constraint is $\mathbb{E}\left\|s_m^{(k)}[l]\right\|^2 = 2|\beta_m(k)|^2 \leq P_0$, where $s_m^{(k)}[l]$ is the transmit signal of device m at the l-th channel use. To compensate the phase shift of channel $h_m(k)$, we construct $\beta_m(k)$ as $\beta_m(k) = b_m(k)e^{-j\theta_m(k)}$, where $\theta_m(k)$ is the phase of $h_m(k)$ and $b_m(k) \in \mathbb{R}$ is amplitude of $\beta_m(k)$. By such a design, the received signal at the UAV-PS, denoted by $\mathbf{y}(k)$, is given by

$$\mathbf{y}(k) = \sum_{m\in\mathcal{M}_k} |h_m(k)|b_m(k)\mathbf{r}_m(k) + \mathbf{n}(k), \quad (14)$$

where $\mathbf{n}(k) \in \mathbb{C}^C$ is an additive white Gaussian noise (AWGN) vector with the elements following the distribution $\mathcal{CN}\left(0, \delta_n^2\right)$. Subsequently, the UAV-PS employs a denoising factor $\zeta(k) \in \mathbb{R}$ to process the received signal $\mathbf{y}(k)$ as

$$\hat{\mathbf{r}}(k) = \zeta(k)\mathbf{y}(k) = \zeta(k)\left(\sum_{m\in\mathcal{M}_k} |h_m(k)|b_m(k)\mathbf{r}_m(k) + \mathbf{n}(k)\right), \quad (15)$$

where $\hat{\mathbf{r}}(k)$ is the processed version of $\mathbf{y}(k)$. Then, the estimate $\hat{\mathbf{g}}(k)$ of desired $\mathbf{g}(k)$ is finally reconstructed as

$$\hat{\mathbf{g}}(k) = \left[\mathrm{Re}\left\{\hat{\mathbf{r}}(k)\right\}^{\mathrm{T}}, \mathrm{Im}\left\{\hat{\mathbf{r}}(k)\right\}^{\mathrm{T}}\right]^{\mathrm{T}} + \bar{g}(k)\mathbf{1}_{D\times 1}, \quad (16)$$

where $\bar{g}(k) = \frac{1}{|\mathcal{M}_k|}\sum_{m\in\mathcal{M}_k}\bar{g}_m(k)$ is introduced to compensate the subtracted mean of gradients $\bar{g}_m(k), m \in \mathcal{M}_k$ in the normalization step (11).

Remark 3: In our UAV-AFL framework, the term "asynchronous" specifically refers to the asynchrony of the learning process, where the gradients used in a single aggregation may be computed based on different versions of the global model across various devices. However, it is important to note that the communication process, particularly the over-the-air aggregation of gradients, is synchronous. For over-the-air aggregation, it is necessary for the gradient data from different devices to be aligned in each dimension during the aggregation process to ensure accurate summation. This alignment requires synchronization among devices in the communication phase.

Algorithm 1 UAV-AFL Framework

Initialization: The UAV-PS distributes the initial model $\{\mathbf{x}(1)\}$ to each device and optimizes $(\mathbf{A}, \mathbf{Q}, \mathbf{B}, \mathbf{Z})$ based on the estimated CSI and device state information.
for $k \in [K]$ **do**
The devices in $\mathcal{M}_k$ transmit $\{\bar{g}_m(k), v_m(k)\}_{m\in\mathcal{M}_k}$ to UAV-PS through error-free links.
The devices in $\mathcal{M}_k$ transmit $\{\mathbf{g}_m(k)\}_{m\in\mathcal{M}_k}$ to the UAV-PS based on (11)-(14).
The UAV-PS recovers the aggregated gradient $\hat{\mathbf{g}}(k)$ based on (15)-(16).
The UAV-PS updates $\mathbf{x}(k)$ based on (10) and broadcasts it to the devices in $\mathcal{M}_k$.
The devices in $\mathcal{M}_k$ carry out computations based on (8) and then wait for the next round of selection.
end for

D. Overall UAV-AFL Framework

We summarize the UAV-enabled over-the-air AFL (UAV-AFL) framework discussed above in Algorithm 1. We define the device selection, UAV-PS trajectory, device transmit amplitude gain and the UAV-PS denoising factor over K time slots as $\mathbf{A} = \{a_m(k) \mid \forall m, k\} \in \mathbb{R}^{M\times K}$, $\mathbf{Q} = \{\mathbf{q}(k) \mid \forall k\} \in \mathbb{R}^{3\times K}$, $\mathbf{B} = \{b_m(k) \mid \forall m, k\} \in \mathbb{R}^{M\times K}$, and $\mathbf{Z} = \{\zeta(k) \mid \forall k\} \in \mathbb{R}^K$, respectively.

IV. Convergence Analysis

In this section, we analyze the convergence of the proposed UAV-AFL system. We present several assumptions in Section IV-A and derive a tractable convergence bound by quantitatively characterizing the impact of $\{\mathbf{A}, \mathbf{Q}, \mathbf{B}, \mathbf{Z}\}$ on the learning efficiency in Section IV-B.

A. Assumptions

To conduct the convergence analysis, we make the following assumptions.

Assumption 1 (Lipschitzian Gradient): The function $F(\cdot)$ is differentiable and the corresponding gradients $\nabla F(\cdot)$ is Lipschitz continuous with parameter L, i.e.,

$$\|\nabla F(\mathbf{x}) - \nabla F(\mathbf{y})\| \leq L\|\mathbf{x} - \mathbf{y}\|, \quad \forall \mathbf{x}, \mathbf{y}. \quad (17)$$

Assumption 2 (Strongly Convex): The function $F(\cdot)$ is strongly convex with parameter μ, i.e.,

$$F(\mathbf{x}) \geq F(\mathbf{y}) + (\mathbf{x} - \mathbf{y})^T\nabla F(\mathbf{y}) + \frac{\mu}{2}\|\mathbf{x} - \mathbf{y}\|^2, \quad \forall \mathbf{x}, \mathbf{y}. \quad (18)$$

Assumption 3 (Staleness Upper Bound): All staleness $\tau_{m,k}, \forall m, k$ are bounded with the value S, i.e., $\tau_{m,k} \leq S, \forall m, k$.

Assumption 4 (Bounded Gradient): The gradient $\nabla F(\mathbf{x}_{k-a_{m,k}})$ with the staleness $a_{m,k} \leq 2S$ is bounded by

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$\nabla F(\mathbf{x}_k)$ with parameters α_1 and α_2, i.e.,

$$\|\nabla F(\mathbf{x}(k-a_{m,k}))\|^2 \le \alpha_1 + \alpha_2 \|\nabla F(\mathbf{x}(k))\|^2, \quad \forall a_{m,k} \le 2S, m, k. \tag{19}$$

Assumption 5 (Device variance): The variance of local gradient and the global counterpart is bounded with parameters δ, i.e.,

$$\|\nabla f_m(\mathbf{x}) - \nabla F(\mathbf{x})\|^2 \le \delta^2, \quad \forall m, k. \tag{20}$$

Assumptions 1, 2 and 5 are widely used in the stochastic optimization literature; see e.g., [24], [28], and [29]. Assumption 1 enables us to choose an appropriate learning rate for balancing convergence speed and learning stability. Assumption 2 ensures the existence of a unique global optimum $\mathbf{x}^\star$ for the loss function $F(\cdot)$. Assumption 5 limits the discrepancy between the gradients of the local and global loss functions.

Assumptions 3 and 4 are introduced because of the asynchronous nature of UAV-AFL. Assumption 3 is commonly employed in the analysis of asynchronous algorithms, e.g., [30] and [31]. Moreover, since the model version gap (staleness) is bounded, we can always find suitable values for α_1 and α_2 ensuring (19) holds.

B. Convergence Analysis

By comparing with the ideal model update case, we rewrite (10) as

$$\mathbf{x}(k+1) = \mathbf{x}(k) - \lambda \hat{\mathbf{g}}(k) = \mathbf{x}(k) - \lambda(\nabla F(\mathbf{x}(k)) - \mathbf{e}(k)) \tag{21}$$

where $\nabla F(\mathbf{x}(k)) = \frac{1}{|\mathcal{S}|}\sum_{\xi_i \in \mathcal{S}} \nabla f(\mathbf{x}(k), \xi_i)$ is the gradient of $F(\mathbf{x})$ at $\mathbf{x} = \mathbf{x}(k)$, and $\mathbf{e}(k)$ represents the overall error introduced during the gradient evaluation process. Under the UAV-AFL framework, $\mathbf{e}(k)$ originates from the following three aspects:

- *Communication error* $\mathbf{e}_c(k)$: The inherent communication noise introduces deviations into aggregated gradient $\hat{\mathbf{g}}(k)$, consequently undermining the learning accuracy.
- *Device selection error* $\mathbf{e}_d(k)$: Given the common case that $\mathcal{M}_k \neq \mathcal{M}$, the aggregated gradients are computed using a subset of data set $\mathcal{S}$, which can potentially result in training imbalance and overfitting [32].
- *Model Asynchrony error* $\mathbf{e}_a(k)$: As shown in (8), the aggregated gradients are evaluated based on different model parameters, so the model inconsistency exists within each aggregation process.

We mathematically formulate the above analysis of $\mathbf{e}(k)$ as

$$\begin{aligned}\mathbf{e}(k) &= \underbrace{\nabla F(\mathbf{x}(k)) - \frac{1}{M}\sum_{m\in\mathcal{M}} \nabla f_m(\mathbf{x}(k-\tau_{m,k}))}_{\mathbf{e}_a(k)} \\ &+ \underbrace{\frac{1}{M}\sum_{m\in\mathcal{M}} \nabla f_m(\mathbf{x}(k-\tau_{m,k})) - \frac{1}{|\mathcal{M}_k|}\sum_{m\in\mathcal{M}_k} \nabla f_m(\mathbf{x}(k-\tau_{m,k}))}_{\mathbf{e}_d(k)} \\ &+ \underbrace{\frac{1}{|\mathcal{M}_k|}\sum_{m\in\mathcal{M}_k} \nabla f_m(\mathbf{x}(k-\tau_{m,k})) - \hat{\mathbf{g}}(k)}_{\mathbf{e}_c(k)}\end{aligned} \tag{22}$$

From (22), we see that errors $\mathbf{e}_c(k)$, $\mathbf{e}_d(k)$ and $\mathbf{e}_a(k)$ collectively compromise the model update process. The overall error $\mathbf{e}(k) = 0$ when the conditions $\tau_{m,k} = 0$, $\mathcal{M}_k = \mathcal{M}$, and $\mathbf{g}(k) = \hat{\mathbf{g}}(k)$ hold simultaneously. We are now ready to analyze the impacts of the errors $\{\mathbf{e}_c(k), \mathbf{e}_d(k), \mathbf{e}_a(k)\}, \forall k$ on the learning efficiency.

Proposition 1: Under Assumptions 1-5, the communication MSE, i.e., $\mathbb{E}\|\mathbf{e}_c(k)\|^2$ is given by (23), the device selection MSE, i.e., $\mathbb{E}\|\mathbf{e}_d(k)\|^2$ is bounded by (24), and the model asynchrony MSE, i.e., $\mathbb{E}\|\mathbf{e}_a(k)\|^2$ is bounded by (25), i.e.,

$$\mathbb{E}\|\mathbf{e}_c(k)\|^2 = \sum_{c=1}^{C} \mathbb{E}\left[\left|\sum_{m\in\mathcal{M}_k} \Delta_m(k) r_m^{(k)}(c) - \zeta(k) n^{(k)}(c)\right|^2\right], \tag{23}$$

$$\mathbb{E}\|\mathbf{e}_d(k)\|^2 \le 8\Theta(k)\left[(M - |\mathcal{M}_k|)/M\right]^2, \tag{24}$$

$$\mathbb{E}\|\mathbf{e}_a(k)\|^2 \le 6S^2\Theta(k) + 6T\mathbb{E}\sum_{j=k-S}^{k-1}\|\mathbf{e}_c(j)\|^2 + 2\delta^2, \tag{25}$$

where $\Delta_m(k) = \frac{\sqrt{v_m(k)}}{|\mathcal{M}_k|} - \zeta(k)|h_m(k)|b_m(k)$, $\Theta(k) = \delta^2 + \alpha_1 + \alpha_2\mathbb{E}\left[\|\nabla F(\mathbf{x}(k))\|^2\right]$.

Proof: Please refer to Appendix A. □

Proposition 1 gives the expression for three errors involved in UAV-AFL system. From Proposition 1, we find that the communication MSE $\mathbb{E}\|\mathbf{e}_c(j)\|^2$ from the previous rounds also have impact on the model asynchrony MSE $\mathbb{E}\|\mathbf{e}_a(k)\|^2$. To obtain a tractable expression of $\mathbb{E}\|\mathbf{e}_c(j)\|^2$, we determine it by leveraging the correlation of local gradients. In this paper, we assume that the local gradients $\{\tilde{\mathbf{g}}_m(k) \mid m \in \mathcal{M}_k\}$ are independent, and the local gradient elements $\{\tilde{g}_m^{(k)}(d) \mid m \in \mathcal{M}_k, \forall d\}$ are independent and identically distributed (i.i.d.). That is,

$$\mathbb{E}\left[\tilde{\mathbf{g}}_i(k)\left(\tilde{\mathbf{g}}_j(k)\right)^{\mathrm{T}}\right] = \mathbf{0}, \quad i \neq j \in \mathcal{M}_k, \tag{26}$$

$$\mathbb{E}\left[\tilde{\mathbf{g}}_m(k)\left(\tilde{\mathbf{g}}_m(k)\right)^{\mathrm{T}}\right] = \mathbf{I}, \quad m \in \mathcal{M}_k. \tag{27}$$

We thus have the following corollary.

Corollary 1: In the UAV-AFL system, under Assumptions 1-5, together with the correlation assumption (26) and (27), the communication MSE $\mathbb{E}\|\mathbf{e}_c(k)\|^2$ is bounded by

$$\begin{aligned}&\mathbb{E}\|\mathbf{e}_c(k)\|^2 \\ &\le \left(\delta^2 + \alpha_1 + \alpha_2\|\nabla F(\mathbf{x}(k))\|^2\right) \\ &\quad\left[|\mathcal{M}_k| - \frac{(\sum_{m\in\mathcal{M}_k}|h_m(k)|b_m(k))^2}{(\sum_{m\in\mathcal{M}_k}|h_m(k)|^2 b_m(k)^2 + \delta_n^2/2)}\right].\end{aligned} \tag{28}$$

Proof: Please refer to Appendix B. □

Based on Corollary 1 and Proposition 1, we have the following Proposition.

Proposition 2: With Assumptions 1-5, $\lambda = 1/L$ and Corollary 1, $\mathbb{E}[F(\mathbf{x}(K+1)) - F(\mathbf{x}^\star)]$ is bounded by

$$\begin{aligned}&\mathbb{E}[F(\mathbf{x}(K+1)) - F(\mathbf{x}^\star)] \\ &\le \prod_{k=1}^{K}\left(1 - \frac{\mu}{L}\left[1 - g(\mathbf{q}_k, \mathbf{a}_k, \mathbf{b}_k)\alpha_2\right]\right)\mathbb{E}[F(\mathbf{x}(1)) - F(\mathbf{x}^\star)]\end{aligned}$$

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$$+\frac{1}{2L}\sum_{k=1}^{K}\left[\left(g(\mathbf{q}_k,\mathbf{a}_k,\mathbf{b}_k)(\delta^2+\alpha_1)+6\delta^2\right)\times\prod_{k'=k+1}^{K}\left(1-\frac{\mu}{L}\left[1-g(\mathbf{q}_{k'},\mathbf{a}_{k'},\mathbf{b}_{k'})\alpha_2\right]\right)\right], \tag{29}$$

where $g(\mathbf{q}_k,\mathbf{a}_k,\mathbf{b}_k) = 18S^2 + 24(\frac{M-\sum_{m=1}^{M}a_m(k)}{M})^2 + 18S\sum_{j=k-S}^{k}(\sum_{m=1}^{M}a_m(j) - \frac{(\sum_{m=1}^{M}a_m(j)\sqrt{g_0}\|\mathbf{q}(j)-\mathbf{w}_m\|^{-1}b_m(j))^2}{\sum_{m=1}^{M}a_m(j)g_0\|\mathbf{q}(j)-\mathbf{w}_m\|^{-2}b_m(j)^2+\delta_n^2/2}).$

Proof: Please refer to Appendix C. □

Proposition 2 characterizes the overall convergence $\mathbb{E}[F(\mathbf{x}(K+1))-F(\mathbf{x}^\star)]$ in UAV-AFL system. We see from Proposition 2 that the overall convergence $\mathbb{E}[F(\mathbf{x}(K+1))-F(\mathbf{x}^\star)]$ is bounded by a quantity depending on UAV trajectory $\mathbf{Q}$, device selection $\mathbf{A}$, and device transmit amplitude gain $\mathbf{B}$.

V. System Optimization

In this section, to improve the learning efficiency of the UAV-AFL system, we conduct a systematic design guided by the analysis in Section IV. Specifically, we establish the objective function based on Proposition 2 and jointly optimize $\{\mathbf{Q},\mathbf{A},\mathbf{B}\}$ to maximize the learning efficiency.

A. Problem Formulation

From Proposition 2, the condition $1-\frac{\mu}{L}[1-g(i)\alpha_2]\leq 1$ must be held to guarantee convergence. Under this condition, as $K\to\infty$, $\mathbb{E}[F(\mathbf{x}(K+1))-F(\mathbf{x}^\star)]$ is dominated by the second term on the right hand side of (29), i.e.,

$$\begin{aligned}&\lim_{K\to\infty}\mathbb{E}[F(\mathbf{x}(K+1))-F(\mathbf{x}^\star)]\\&\leq\sum_{k=1}^{K}\frac{1}{2L}\left[g(\mathbf{q}_k,\mathbf{a}_k,\mathbf{b}_k)(\delta^2+\alpha_1)+6\delta^2\right]\\&\quad\times\prod_{k'=k+1}^{K}\left(1-\frac{\mu}{L}\left[1-g(\mathbf{q}_{k'},\mathbf{a}_{k'},\mathbf{b}_{k'})\alpha_2\right]\right).\end{aligned} \tag{30}$$

With the objective in (30), we formulate the UAV-AFL communication-learning design problem over $\{\mathbf{Q},\mathbf{A},\mathbf{B}\}$ as

$$\min_{\mathbf{Q},\mathbf{A},\mathbf{B}}\ \sum_{k=1}^{K}\frac{1}{2L}\left[g(\mathbf{q}_k,\mathbf{a}_k,\mathbf{b}_k)(\delta^2+\alpha_1)+6\delta^2\right]\times\prod_{k'=k+1}^{K}\left(1-\frac{\mu}{L}\left[1-g(\mathbf{q}'_k,\mathbf{a}'_k,\mathbf{b}'_k)\alpha_2\right]\right) \tag{31a}$$

$$\text{s.t.}\quad\sum_{k}^{k+S-1}a_m(k)\geq 1,\quad\forall m,k\leq K-S+1, \tag{31b}$$

$$\sum_{k}^{k+c_m}a_m(k)\leq 1,\quad\forall m,k\leq K-c_m, \tag{31c}$$

$$a_m(k)\in\{0,1\},\quad\forall m,k \tag{31d}$$

$$b_m(k)^2\leq P_0/2,\quad\forall m,(4)-(6). \tag{31e}$$

where (31d) denotes the binary constraints for the device selection variable $\mathbf{A}$; (31b) is the staleness constraint from Assumption 3, which is introduced to mitigate the impact of stale models on the learning process; (31c) represents that the devices cannot be selected during the local computing process; $b_m(k)^2\leq P_0/2$ is the device transmit power constraint; (4)-(6) are the mechanical constraints of the UAV-PS.

Remark 4: Problem (31) is a mixed-integer programming problem. To address this, we use geometric programming (GP) to transform the objective function (31a) into a tractable convex form. Then, an efficient two-layer penalty based algorithm using alternative optimization (AO) and successive convex approximation (SCA) is developed to solve this problem, as discussed in what follows.

B. Problem Transform via Geometric Programming

To deal with the non-convex objective function, we introduce auxiliary variables $\mathbf{p}\in\mathbb{R}^{2K}$ and $\mathbf{y}\in\mathbb{R}^{2K}$ to recast problem (31) as

$$\min_{\mathbf{Q},\mathbf{A},\mathbf{B},\mathbf{p},\mathbf{y}}\ \log\left(\sum_{k=1}^{K}e^{\mathbf{c}_k^{\mathrm{T}}\mathbf{y}}\right) \tag{32a}$$

$$\text{s.t.}\quad\frac{\delta^2+\alpha_1}{2L}g(\mathbf{q}_k,\mathbf{a}_k,\mathbf{b}_k)+3\frac{\delta^2}{L}\leq p_k,\quad\forall k, \tag{32b}$$

$$1-\frac{\mu}{L}+\frac{\mu}{L}g(\mathbf{q}_k,\mathbf{a}_k,\mathbf{b}_k)\alpha_2\leq p_{K+k},\quad\forall k, \tag{32c}$$

$$p_i\leq e^{y_i},\quad\forall i \tag{32d}$$

$$(31d)-(31e), \tag{32e}$$

where $\mathbf{c}_k\in\mathbb{R}^{2K}$ with $c_k^k=c_k^{K+k+1}=c_k^{K+k+2}=\cdots=c_k^{2K}=1$ and other elements of $\mathbf{c}_k$ are 0.

Proposition 3: Problem (32) is an equivalent form of (31).

Proof: The auxiliary variable $\mathbf{p}$ satisfying (32b) and (32c) is used to recast (31) as

$$\min_{\mathbf{Q},\mathbf{A},\mathbf{B},\mathbf{p}}\ \sum_{k=1}^{K}p_k\prod_{k'=k+1}^{K}p_{K+k'} \tag{33a}$$

$$\text{s.t.}\quad(32b),(32c),(31d)-(31e). \tag{33b}$$

Note that the objective (33a) monotonically increases with respect to $\mathbf{p}$. Hence, the optimal $\mathbf{p}$ must be obtained at the lower boundary of the feasible region of problem (33). Consequently, at the optimum of (33), constraints (32b) and (32c) are satisfied with equality. This means the equivalence of problems (33) and (31). Moreover, the equivalence between (33) and (32) can be proved similarly by considering auxiliary variable $\mathbf{y}$. The proof is thus complete. □

C. Penalty Based Binary Constraint

The conventional method [33], [34] for tackling the binary device selection constraint (31d) is to convert the binary elements of $\mathbf{A}$ to continuous-valued variables in optimization and then rounding the optimized $\mathbf{A}$. However, the obtained $\mathbf{A}$ by this way may not satisfy the staleness constraint (31b) and the computing time constraint (31c). In this paper, a penalty based method is used to handle this constraint. We initially introduce auxiliary variables to transform (31b) into a series of equality constraints. Subsequently, to ensure the satisfaction of these equality constraints, we incorporate them into the objective

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function in the form of penalty terms. Specifically, we introduce $\bar{\mathbf{A}} = \{\bar{a}_m(k), \forall m, k\} \in \mathbb{R}^{M\times K}$ and rewrite (31d) as

$$a_m(k)(1-\bar{a}_m(k)) = 0, \quad a_m(k) = \bar{a}_m(k), \ \forall m, k. \tag{34}$$

Clearly, the elements of $\mathbf{A}$ must be either 0 or 1 to comply with the constraints (34). We then introduce penalty terms in the form of (34) into the objective function [35], yielding the following formulation:

$$\min_{\mathbf{Q},\mathbf{A},\bar{\mathbf{A}},\mathbf{B},\mathbf{p},\mathbf{y}} \quad \log\left(\sum_{k=1}^{K} e^{\mathbf{c}_k^{\mathrm{T}}\mathbf{y}}\right) + \frac{1}{\eta}\sum_{m=1}^{M} P(\mathbf{A},\bar{\mathbf{A}}) \tag{35a}$$

$$\text{s.t.} \quad (32b), (32c), (32d), (31b) - (31e), \tag{35b}$$

where $P(\mathbf{A},\bar{\mathbf{A}}) = [a_m(k)(1-\bar{a}_m(k))]^2 + [a_m(k) - \bar{a}_m(k)]^2$, and $\eta > 0$ is employed to penalize any deviation from the equality constraint (34). It can be easily verified that the binary constraint (31d) must be satisfied as $\frac{1}{\eta} \to \infty$.

D. Inner and Outer Layer Iteration

In this subsection, we propose a two-layer iterative algorithm. Specifically, in the inner layer, we decompose (35) into three sub-problems, wherein the variables $\bar{\mathbf{A}}$, $\mathbf{A}$ and $\{\mathbf{Q},\mathbf{B}\}$ are optimized separately. In the outer layer, we gradually decrease the coefficient η. The details are shown as follows.

1) Slack Variable Optimization: We first optimize slack variable $\bar{\mathbf{A}}$ with given $\{\mathbf{A},\mathbf{Q},\mathbf{B}\}$. By dropping the irrelevant terms, $\bar{\mathbf{A}}$ appears exclusively in the objective function as a convex form, and the optimal value of $\bar{\mathbf{A}}$ is

$$\bar{a}_m^{\star}(k) = \frac{a_m^2(k) + a_m(k)}{1 + a_m^2(k)} \tag{36}$$

2) Device Selection Optimization: We now optimize the device selection variable $\mathbf{A}$. With given $\{\bar{\mathbf{A}},\mathbf{Q},\mathbf{B}\}$, problem (35) can be reformulated as

$$\min_{\mathbf{A},\mathbf{p},\mathbf{y}} \quad \log\left(\sum_{k=1}^{K} e^{\mathbf{c}_k^{\mathrm{T}}\mathbf{y}}\right) + \frac{1}{\eta}\sum_{m=1}^{M} P(\mathbf{A},\bar{\mathbf{A}}) \tag{37a}$$

$$\text{s.t.} \quad g(\mathbf{a}_k) \le \Lambda(k), \quad \forall k \tag{37b}$$

$$(31b), (31c), (32d), \tag{37c}$$

where $\Lambda(k) = \min\left(\frac{2Lp_k - 6\delta^2}{\delta^2+\alpha_1}, \frac{Lp_{K+k}+\mu-L}{\mu\alpha_2}\right)$. The problem (37) is neither convex or quasi-convex because of the non-convex constraints (37b) and (32d). Here, we introduce auxiliary variables $\{\mathbf{f},\mathbf{d}\}$ and use the successive convex approximation (SCA) technique to transform (37) to a convex form as

$$\min_{\mathbf{A},\mathbf{p},\mathbf{y},\mathbf{f},\mathbf{d}} \quad \log\left(\sum_{k=1}^{K} e^{\mathbf{c}_k^{\mathrm{T}}\mathbf{y}}\right) + \frac{1}{\eta}\sum_{m=1}^{M} P(\mathbf{A},\bar{\mathbf{A}}) \tag{38a}$$

$$\text{s.t.} \quad f_k \le \Lambda(k), \quad \forall k, \tag{38b}$$

$$p_i \le e^{y_i^t} + e^{y_i^t}(y_i - y_i^t), \quad \forall i, \tag{38c}$$

$$f_k \ge 18S^2 + \frac{24}{M^2}\Phi(k)^2 + 18S\Psi(k), \quad \forall k, \tag{38d}$$

$$d_k \le R(k) + (\mathbf{a}_k - \mathbf{a}^t(k))^{\mathrm{T}}(\Upsilon(k) + \Xi(k)), \quad \forall k, \tag{38e}$$

$$(31b), (31c), \tag{38f}$$

Algorithm 2 Two-Layer Iterative Algorithm

1: **Initialize:** The initial value $\bar{\mathbf{A}}^0, \mathbf{Q}^0, \mathbf{A}^0$ and $\mathbf{B}^0$, iteration index $t = 1$, predefined precision ϵ_1 and ϵ_2.
2: **repeat**
3: **repeat**
4: With given $\{\mathbf{Q}^t, \mathbf{A}^t, \mathbf{B}^t\}$, compute $\bar{\mathbf{A}}^{t+1}$ based on (36).
5: With given $\{\bar{\mathbf{A}}^t, \mathbf{Q}^t, \mathbf{B}^t\}$, compute $\mathbf{A}^{t+1}$ by solving problem (38).
6: With given $\{\bar{\mathbf{A}}^t, \mathbf{A}^t\}$, compute $\{\mathbf{Q}^{t+1}, \mathbf{B}^{t+1}\}$ by solving the convexified problem.
7: Calculate the value L^{t+1} of objective function in (35).
8: $t = t + 1$.
9: **until** $\left|(L^t - L^{t-1})/L^{t-1}\right| \le \epsilon_1$
10: $\eta = s\eta$.
11: **until** the violation of (34) falls below the threshold ϵ_2.
Output: $\{\mathbf{Q},\mathbf{A},\mathbf{B}\}$.

where $\{\mathbf{A}^t, \mathbf{y}^t\}$ are the optimized results of $\{\mathbf{A},\mathbf{y}\}$ obtained at the previous inner iteration, $\mathbf{a}^t(k)$ is the k-th column of $\mathbf{A}^t$, $\Psi(k) = \sum_{j=k-S}^{k}(\sum_{m=1}^{M} a_m(j) - d(j))$, $\Phi(k) = M - \sum_{m=1}^{M} a_m(k)$, $\mathbf{v}(k) = [|h_1(k)|b_1(k), \cdots, |h_M(k)|b_M(k)]$, $R(k) = \frac{\mathbf{a}^t(k)^{\mathrm{T}}\mathbf{v}(k)\mathbf{v}(k)^{\mathrm{T}}\mathbf{a}^t(k)}{\mathbf{a}^t(k)^{\mathrm{T}}\mathbf{v}(k)\circ\mathbf{v}(k)+\delta_n^2/2}$, $\Omega(k) = \frac{2\mathbf{v}(k)\mathbf{v}(k)^{\mathrm{T}}\mathbf{a}^t(k)}{(\mathbf{a}^t(k)^{\mathrm{T}}\mathbf{v}(k)\circ\mathbf{v}(k)+\delta_n^2/2)^2}$, $\Upsilon(k) = \Omega(k)\cdot(\mathbf{a}^t(k)^{\mathrm{T}}\mathbf{v}(k)\circ\mathbf{v}(k) + \delta_n^2/2)$, $\Xi(k) = -\frac{\mathbf{a}^t(k)^{\mathrm{T}}\mathbf{v}(k)\mathbf{a}^t(k)^{\mathrm{T}}\mathbf{v}(k)\mathbf{v}(k)\circ\mathbf{v}(k)}{(\mathbf{a}^t(k)^{\mathrm{T}}\mathbf{v}(k)\circ\mathbf{v}(k)+\delta_n^2/2)^2}$.

Proposition 4: By solving problem (38), we obtain a locally optimal solution of (37) satisfying the Karush-Kuhn-Tucker (KKT) condition, which serves as an upper bound of (37a).

Proof: Please refer to Appendix D. □

3) Trajectory and Transmit Power Optimization: Similar to the discussions of (37), with fixed $\{\mathbf{A},\bar{\mathbf{A}}\}$, (35) can be solved by using SCA to obtain an update of $\{\mathbf{Q},\mathbf{B}\}$. Due to page limitations, we omit the details here.

4) Outer Layer Update: In the outer layer, the penalty coefficient η is iteratively updated by $\eta = s\eta$, where s represents a scaling factor with range $0 < s < 1$.

E. Overall Algorithm and Complexity Analysis

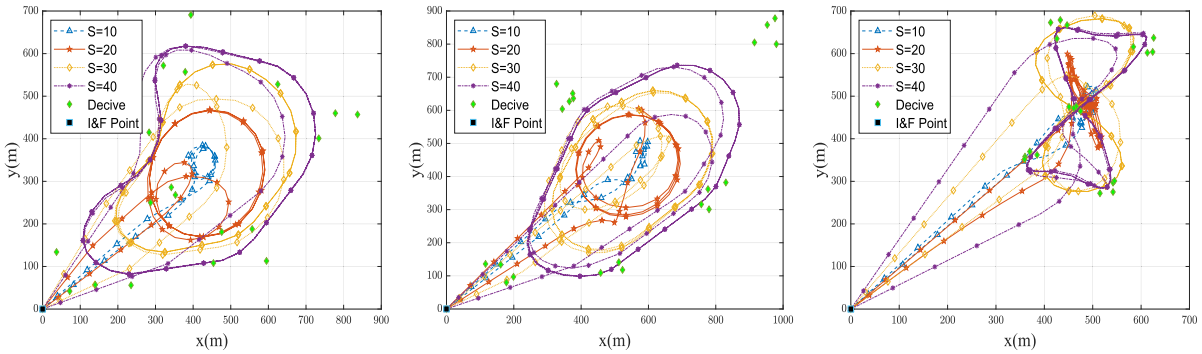
We summarize the proposed algorithm in Algorithm 2.

We now analyze the complexity of Algorithm 2. The computational complexity of Algorithm 2 is dominated by steps 5 and 6. We employ the convex optimization tool CVX [36], which utilizes the standard interior-point method, to address the problems posed in these steps. Consequently, the complexity involved in steps 5 and 6 is $\mathcal{O}((MK)^{3.5})$. Therefore, the overall complexity of Algorithm 2 is $\mathcal{O}(I_{outer}I_{inner}(MK)^{3.5})$, where I_{outer} and I_{inner} denote the numbers of iterations to achieve the precision in the inner and outer layers, respectively.

VI. Numerical Result

In this section, we present numerical results to evaluate the performance of the proposed UAV-AFL scheme. These results

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(a) Trajectory of Scenario 1. (b) Trajectory of Scenario 2. (c) Trajectory of Scenario 3.

Fig. 3. Trajectory of different staleness bounds in three scenarios.

offer valuable insights for the UAV-AFL implementation and demonstrate the superior performance of the proposed scheme.

A. Simulation Setup

Unless otherwise specified, the system parameters are configured as follows. We consider an area of 1 km $\times$ 1 km with a total of $M = 20$ edge devices. The maximum flight speed $v_{\max}$ (or acceleration $a_{\max}$) of the UAV-PS is 50 m/s (or 15 m/s^2), and the fixed flying altitude $H = 100$ m [11]. We set the initial and final dispatch point of UAV-PS as $\mathbf{q}_0 = \mathbf{q}_F(0, 0, 100)$; the reference channel gain $g_0 = -60$ dB; the noise power $\sigma_n^2 = -80$ dBm; the maximum transmit power of devices $P_0 = 0.1$ W [12]; the staleness bound $S = 50$, the number of overall time slots $K = 10000$; the duration of each time slot $\delta_t = 1$ s; the threshold ϵ_1 (or ϵ_2) is 10^{-2} (or 1). For learning parameters, we set $\mu = 0.2$, $L = 10$, $\delta = 20$, $\alpha_1 = 100$ and $\alpha_2 = 0.1$ [37]. To provide a clear exhibition of the UAV-PS trajectory, we chose $K = 250$ to obtain the optimization results, and use this as a basic cycle to conduct the UAV-AFL system training periodically for 10000 time slots.

For the learning configuration, we conduct image classification task using the MNIST dataset. We train a convolutional neural network with two 5×5 convolutional layers (each followed by 2×2 max pooling), along with a batch normalization layer, a fully connected layer consisting of 50 units, a ReLu activation layer, and finally a softmax output layer. The loss function employed is the cross-entropy loss. The local data of edge devices are i.i.d. and equally drawn from a pool of 5×10^4 images while the test data set contains 10^4 images. The experiments and simulations are conducted using Python with an NVIDIA RTX 3060Ti GPU.

B. Comparisons Based on Different Staleness Bound

In this subsection, we investigate the impact of different staleness bounds S on the performance of the UAV-AFL system. We conduct simulations in three scenarios. In scenario 1, the devices are randomly located. In scenario 2 and 3, the devices are located within different clusters. We use the metric $\sigma(\mathbf{w}) \triangleq \sum_{m=1}^{M} \|\mathbf{w}_m - \overline{\mathbf{w}}_m\|$ to roughly measure the standard deviation of devices' geographical distributions. We introduce the communication normalized mean square error (NMSE) metric, i.e., $10 \log_{10} \left(\mathbb{E} \left[\|\hat{\mathbf{g}}(k) - \mathbf{g}(k)\|^2 \right] / \|\mathbf{g}(k)\|^2 \right)$ to quantify the communication quality of time slot k. Table II shows the average of the communication NMSE under K time slots. All the presented results are averaged over 50 Monte Carlo trials.

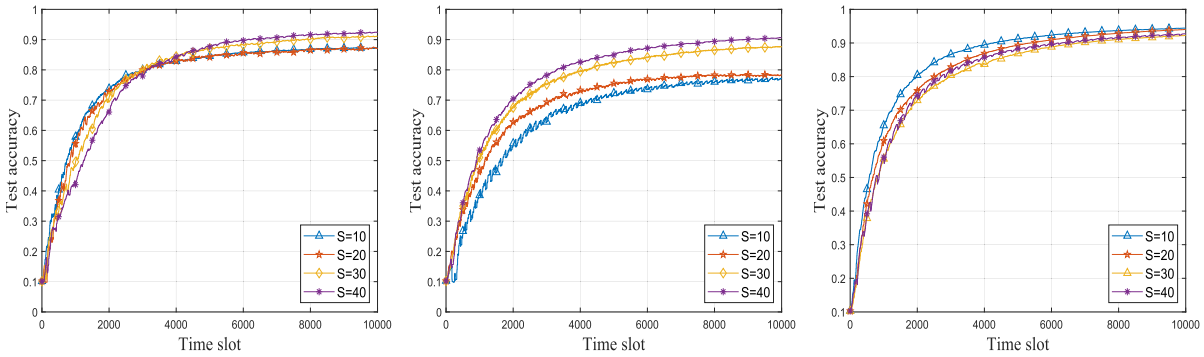
TABLE II
AVERAGE COMMUNICATION NMSE

	NMSE (dB)		
	Scenario 1	Scenario 2	Scenario 3
S=10	-5.3743	-1.4945	-12.0915
S=20	-7.3219	-3.0754	-12.7055
S=30	-8.3004	-6.8537	-14.3379
S=40	-10.4780	-9.4066	-16.9662

From Fig. 3, it is evident that a larger staleness bound provides UAV-PS more time to approach devices for gradient collection. This results in more expansive flight trajectories and reduced communication NMSE, as shown in Table II. Conversely, with a more stringent staleness constraint, the UAV-PS is compelled to complete a service cycle more rapidly, resulting in narrower trajectories. In the simulations, we note the UAV-PS stops communicating with devices under suboptimal channel conditions, especially when there is no proximal device. This behavior underscores the efficacy of our proposed algorithm in balancing device selection with communication quality, where the UAV-PS only aggregates the gradients when communication quality is favorable.

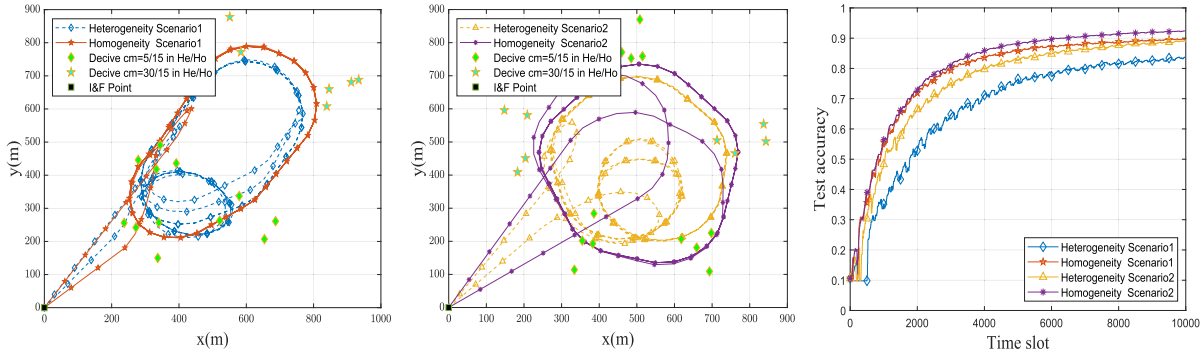
In Fig. 4, the learning performance varies distinctly across different staleness upper bounds S. For devices located in close proximity (scenario 3, with $\sigma(\mathbf{w}) = 153.36$ m), the UAV-PS can have favorable channel quality without the necessity of nearing each device. Under such circumstances, smaller S values exhibit higher learning efficiency, attributable to shorter service cycles and faster model update rate. For scenario 2 where devices' location are more dispersed ($\sigma(\mathbf{w}) = 384.53$ m), communication quality becomes the main challenge. Here, a larger S (i.e., $S = 30, 40$) allows the UAV-PS the enough time to approach devices and enhance the quality of transmission, leading to improved performance. Lastly, scenario 1, with $\sigma(\mathbf{w}) = 269.75$ m, represents an intermediate case between scenarios 2 and 3. In this scenario, at the early stages of training ($k < 4000$), the learning accuracy is predominantly impacted by the model update rate, making

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(a) Performance of Scenario 1. (b) Performance of Scenario 2. (c) Performance of Scenario 3.

Fig. 4. Performance of different staleness bounds in three scenarios.



(a) Trajectories of scenario 1. (b) Trajectories of scenario 2. (c) Performance of scenario 1 and 2.

Fig. 5. System performance under heterogeneous and homogeneous settings.

schemes with a smaller S superior. However, as training progresses ($k > 4000$), the adverse effects of communication errors become pronounced (as the model stabilizes). In this phase, schemes with a larger S exhibit better test accuracy.

C. Performance Under Heterogeneous Computing Ability

In this subsection, we conduct experiments under the FL system where edge devices have heterogeneous computing abilities. In the simulations, devices are grouped into five distinct clusters based on their locations, where the devices within the same cluster are set to have the same gradient computation time c_m defined in Section III-B.

In Fig. 5, we study the performance of UAV-AFL scheme under two scenarios with different device distributions. In each scenario, both heterogeneous and homogeneous configurations are considered. For the heterogeneous setting, the computation time c_m for devices within three clusters (cyan diamond icons) is 5 s while the computation time for devices in other clusters (green pentagon icons) is 30 s. For the homogeneous setting, the computation time c_m for each device is set to 15 s.[2]

We see from Fig. 5 (a) and (b) that the UAV-PS alters its flight strategy in the two different settings significantly. In the homogeneous setting, the UAV-PS appears to traverse and poll each device cluster in sequence. Conversely, in the heterogeneous setting, the UAV-PS prioritizes devices with superior computational abilities. For example, in scenario 1, devices with superior computational capabilities ($c_m = 5$ s) are selected, on average, 1.92 times more frequently than less capable devices ($c_m = 30$ s). Hence, the proposed flight strategy can leverage the gradient computation time to facilitate the model updates.

In Fig. 5 (c), we note that the heterogeneity adversely impacts the UAV-AFL learning performance. This can be understood from two perspectives. Firstly, the heterogeneity prompts the UAV-PS to rely heavily on devices with superior computational capabilities for model updates, which introduces a bias in the learned model. Secondly, heterogeneous settings often result in larger staleness degree compared to homogeneous settings. We quantify this by calculating the maximum staleness (MS), defined as $\max\{\tau_{m,k} \mid \forall m, \forall k\}$, within our simulation. This can be considered as the tightest staleness bound S achieve by the practical UAV-AFL system. For example, in Scenario (a), the MS values for heterogeneous and homogeneous configurations are 40 and 28, respectively. As suggested by Proposition 2, the learning performance is compromised by the large value of MS.

D. Performance Comparison With Benchmarks

In this subsection, we present a comparison of the proposed scheme with several existing state-of-the-art solutions to demonstrate the superior UAV-AFL performance. The benchmarks are as follows.

- **Error free channel (Error free)**: We use the device schedule result i.e., $\mathbf{A}$ obtained by the Algorithm 2 to update the global model without communication error, i.e., $\hat{\mathbf{g}}(k) = \mathbf{g}(k)$.

[2]By this setting, the total time for all devices to complete computation are the same under both heterogeneous and homogeneous settings.

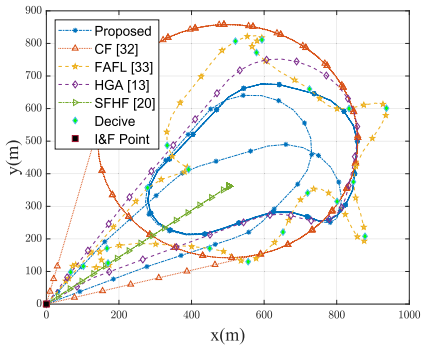
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- **Circular flight trajectory (CF) [38]**: The UAV-PS flies with a circular trajectory [38] centered at $(500, 500, H)$ using the maximum velocity. The radius of trajectory R is determined heuristically by the average distance minimization problem, i.e., $\min_R \sum_{i=1}^{M}(R - \sqrt{x_m^2 + y_m^2})^2$. Given the trajectory, the device selection and the transmit amplitude gain are optimized by Algorithm 2.
- **Synchronous Fly-Hover-Fly (SFHF) [25]**: The UAV-PS proceeds directly from its initial position to a optimized location. At this location, the UAV-PS serves as a stationary central PS providing synchronous FL update [25] where all devices upload the gradients over-the-air. The location $\mathbf{q}$ is determined by

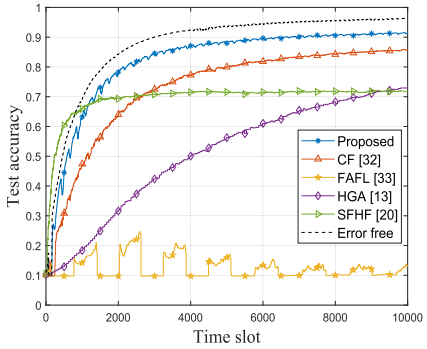
$$\min_{\mathbf{q},\mathbf{b}} \quad M - \frac{(\sum_{m\in\mathcal{M}} \sqrt{g_0 \left\|\mathbf{q} - \mathbf{w}_m\right\|^{-2}} b_m)^2}{\sum_{m\in\mathcal{M}} g_0 \left\|\mathbf{q} - \mathbf{w}_m\right\|^{-2} b_m^2 + \delta_n^2/2}. \quad (39)$$

- **Hierarchical gradient aggregation (HGA) [17]**: The service strategy in [17] is simulated. Under this approach, the UAV-PS is programmed to aggregate the gradients of devices located within a threshold of $d_{\text{thr}} = 250$ m from the UAV-PS over-the-air. After all the gradients are collected, the UAV-PS further combines the previous aggregated gradients and updates the global model.
- **Fully asynchronous FL with error-free communication (FAFL) [39]**: Following the fully asynchronous FL principle [39], only a singular edge device is permitted to upload its gradient during one time slot. In this paradigm, the UAV-PS is mandated to sequentially approach each device to ensure effective gradient transmission. Given that each transmission is under a point-to-point communication link with low path loss, it is reasonable to assume error-free communication. The technique for solving the traveling salesman problem (TSP) is employed to optimize the UAV-PS trajectory in this scheme.

In Fig. 6, we depict the trajectory and the test accuracy of various schemes in a scenario where devices are dispersedly located ($\sigma(\mathbf{w}) = 340.82$ m). It is evident that the proposed UAV-AFL scheme offers a substantial performance enhancement in comparison to the benchmarks, and is similar to the error-free case. The disparities in learning accuracy across different schemes can be attributed to the characteristics of their service strategies. Specifically, although the accuracy of SFHF sees a rapid rise at the beginning (due to the use of all devices' gradients for updates), it becomes stabilized at a less-than-optimal level (due to the restriction by the poor communication quality). Moreover, because of the fully asynchronous nature of FAFL, this scheme suffers from a pronounced staleness degree (MS $= 114$). Hence, its accuracy oscillates around 0.1 throughout the training period. Furthermore, the learning accuracy under the HGA method experiences a slowly ascent throughout this period. This is because the hierarchical strategy necessitates the waiting for all devices' gradient collection, which inevitably impair the learning efficiency. Lastly, the inefficient performance of CF approach highlights the importance of trajectory optimization. For scenarios with limited computational resources for optimization, the CF scheme can be a competitive alternative.

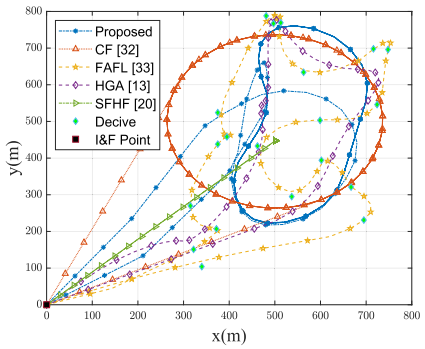


(a) Trajectories

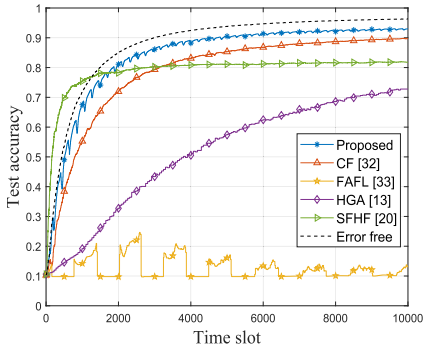


(b) Test accuracies

Fig. 6. Performance comparison of different schemes in scenario 1.



(a) Trajectories



(b) Test accuracies

Fig. 7. Performance comparison of different schemes in scenario 2.

In Fig. 7, we present the trajectory and the test accuracy in a scenario where the devices are densely located ($\sigma(\mathbf{w}) = 213.31$ m). The results are similar with those depicted in Fig. 6, which demonstrates the robust performance gain achieved by the proposed scheme.

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VII. Conclusion

In this paper, we investigated the design of the UAV-AFL system over edge networks. We proposed a novel instant-update service strategy to improve the learning efficiency. We introduced a staleness bound metric to limit the level of AFL model asynchrony and derived an overall convergence bound by capturing the impact of model asynchrony, device selection and communication errors on the UAV-AFL learning performance. This bound represents a systematic endeavor to evaluate AFL performance under the consideration of both communication and learning perspectives. Subsequently, an optimization problem is then formulated to jointly determine UAV-PS trajectory, device selection and over-the-air transceiver design. To address this, we proposed a two-layer iterative algorithm leveraging geometric programming and successive convex approximation for efficient problem-solving. Our numerical results demonstrated the superior effectiveness of the proposed UAV-AFL scheme when benchmarked against state-of-the-art approaches.

Appendix A
Proof of Proposition 1

For notation convenience, we denote $\nabla f_m(\mathbf{x}(k-\tau_{m,k}))$ and $\nabla F(\mathbf{x}(k-\tau_{m,k}))$ as $\nabla f_{m,k}$ and $\nabla F_{m,k}$, respectively. Firstly, the communication MSE $\|\mathbf{e}_c(k)\|^2$ is given by

$$\mathbb{E}\left[\|\mathbf{e}_c(k)\|^2\right] = \mathbb{E}\left[\left\|\frac{1}{|\mathcal{M}_k|}\sum_{m\in\mathcal{M}_k}\nabla f_{m,k} - \hat{\mathbf{g}}(k)\right\|^2\right]$$

Plugging (8), (9), (11), (12), (15) and (16) into the above equality, we obtain (23).

We use $\mathcal{P}_k$ as the complement of $\mathcal{M}_k$, i.e., $\mathcal{P}_k \cup \mathcal{M}_k = \mathcal{M}$. For device selection MSE $\|\mathbf{e}_d(k)\|^2$, we have

$$\begin{aligned}
&\mathbb{E}\left[\|\mathbf{e}_d(k)\|^2\right]\\
&= \mathbb{E}\Big[\Big\|\frac{1}{M}\sum_{m\in\mathcal{M}}\nabla f_{m,k} - \frac{1}{|\mathcal{M}_k|}\sum_{m\in\mathcal{M}_k}\nabla f_{m,k}\Big\|^2\Big]\\
&= \mathbb{E}\Big[\frac{M-|\mathcal{M}_k|}{M|\mathcal{M}_k|}\Big(\Big\|\sum_{m\in\mathcal{M}_k}(\nabla f_{m,k}-\nabla F_{m,k})\Big\|\\
&\quad + \Big\|\sum_{m\in\mathcal{M}_k}\nabla F_{m,k}\Big\|\Big)\\
&\quad + \frac{1}{M}\Big\|\sum_{m\in\mathcal{P}_k}(\nabla f_{m,k}-\nabla F_{m,k})\Big\| + \frac{1}{M}\Big\|\sum_{m\in\mathcal{P}_k}\nabla F_{m,k}\Big\|\Big]^2\\
&\overset{(a)}{\le} \mathbb{E}\Big[\frac{2(M-|\mathcal{M}_k|)}{M}\delta + \frac{M-|\mathcal{M}_k|}{M|\mathcal{M}_k|}\Big\|\sum_{m\in\mathcal{M}_k}\nabla F_{m,k}\Big\|\\
&\quad + \frac{1}{M}\Big\|\sum_{m\in\mathcal{P}_k}\nabla F_{m,k}\Big\|\Big]^2\\
&\overset{(b)}{\le} 4\mathbb{E}\Big[\frac{M-|\mathcal{M}_k|}{M}\delta + \frac{M-|\mathcal{M}_k|}{M}\sqrt{\alpha_1+\alpha_2\|\nabla F(\mathbf{x}(k))\|^2}\Big]^2\\
&\overset{(c)}{\le} 8\mathbb{E}\Big(\frac{M-|\mathcal{M}_k|}{M}\Big)^2\left(\delta^2+\alpha_1+\alpha_2\|\nabla F(\mathbf{x}(k))\|^2\right),
\end{aligned}$$

where (a) is from triangle inequality and Assumption 5, (b) is from Assumption 4, and (c) is from the triangle inequality.

Finally, we bound the model asynchrony $\|\mathbf{e}_a(k)\|^2$.

$$\begin{aligned}
\mathbb{E}\|\mathbf{e}_a(k)\|^2 &= \mathbb{E}\Big[\Big\|\nabla F(\mathbf{x}(k)) - \frac{1}{M}\sum_{m=1}^{M}\nabla f_{m,k}\Big\|^2\Big]\\
&\le \frac{1}{M}\sum_{m=1}^{M}\mathbb{E}\left[\|(\nabla F(\mathbf{x}(k)) - \nabla f_{m,k})\|^2\right]\\
&\overset{(a)}{\le} \frac{2}{M}\sum_{m=1}^{M}\mathbb{E}\left[\|\nabla F(\mathbf{x}(k)) - \nabla F_{m,k}\|^2\right.\\
&\quad \left. + \|\nabla F_{m,k} - \nabla f_{m,k}\|^2\right]\\
&\overset{(b)}{\le} \frac{2}{M}\sum_{m=1}^{M}\mathbb{E}\|\nabla F(\mathbf{x}(k)) - \nabla F_{m,k}\|^2 + 2\delta^2\\
&\overset{(c)}{\le} \frac{2L^2}{M}\sum_{m=1}^{M}\mathbb{E}\|\mathbf{x}(k) - \mathbf{x}(k-\tau_{m,k})\|^2 + 2\delta^2\\
&\overset{(d)}{\le} 2L^2\mathbb{E}\|\mathbf{x}(k) - \mathbf{x}(k-\tau_{k,\mu})\|^2 + 2\delta^2,
\end{aligned}$$

where (a) is from the triangle inequality, (b) is from Assumption 5, (c) is from Assumption 1. From Assumption 3, since the staleness is bounded, we can always find a variable μ satisfying $\mu = \arg\max_{m\in[M]}\|x(k) - \mathbf{x}(k-\tau_{m,k})\|^2$ and hence the step (d) holds. It then follows that

$$\begin{aligned}
&L^2\mathbb{E}\|\mathbf{x}(k) - \mathbf{x}(k-\tau_{k,\mu})\|^2\\
&= L^2\mathbb{E}\Big\|\sum_{j=k-\tau_{k,\mu}}^{k-1}(\mathbf{x}(j+1) - \mathbf{x}(j))\Big\|^2\\
&= \mathbb{E}\Big\|\sum_{j=k-\tau_{k,\mu}}^{k-1}\Big(\hat{\mathbf{g}}(j) - \frac{1}{|\mathcal{M}_j|}\sum_{m\in\mathcal{M}_j}\nabla F_{m,j}\Big)\\
&\quad + \sum_{j=k-\tau_{k,\mu}}^{k-1}\frac{1}{|\mathcal{M}_j|}\sum_{m\in\mathcal{M}_j}\nabla F_{m,j}\Big\|^2\\
&= \mathbb{E}\Big\|\sum_{j=k-\tau_{k,\mu}}^{k-1}\frac{1}{|\mathcal{M}_j|}\sum_{m\in\mathcal{M}_j}(\nabla f_{m,j} - \nabla F_{m,j})\\
&\quad + \sum_{j=k-\tau_{k,\mu}}^{k-1}\frac{1}{|\mathcal{M}_j|}\sum_{m\in\mathcal{M}_j}\nabla F_{m,j} - \sum_{j=k-\tau_{k,\mu}}^{k-1}\mathbf{e}_c(j)\Big\|^2\\
&\overset{(a)}{\le} 3\mathbb{E}\Big\|\sum_{j=k-\tau_{k,\mu}}^{k-1}\frac{1}{|\mathcal{M}_j|}\sum_{m\in\mathcal{M}_j}(\nabla f_{m,j} - \nabla F_{m,j})\Big\|^2\\
&\quad + 3\mathbb{E}\Big\|\sum_{j=k-\tau_{k,\mu}}^{k-1}\frac{1}{|\mathcal{M}_j|}\sum_{m\in\mathcal{M}_j}\nabla F_{m,j}\Big\|^2\\
&\quad + 3\mathbb{E}\Big\|\sum_{j=k-\tau_{k,\mu}}^{k-1}\mathbf{e}_c(j)\Big\|^2\\
&\overset{(b)}{\le} 3S\mathbb{E}\sum_{j=k-S}^{k-1}\frac{1}{|\mathcal{M}_j|}\sum_{m\in\mathcal{M}_j}\|\nabla f_{m,j} - \nabla F_{m,j}\|^2\\
&\quad + 3S\mathbb{E}\sum_{j=k-S}^{k-1}\frac{1}{|\mathcal{M}_j|^2}\Big\|\sum_{m\in\mathcal{M}_j}\nabla F_{m,j}\Big\|^2\\
&\quad + 3S\sum_{j=k-S}^{k-1}\mathbb{E}\|\mathbf{e}_c(j)\|^2
\end{aligned}$$

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$$\overset{(c)}{\leq} 3S^2\delta^2 + 3T\mathbb{E}\sum_{j=k-S}^{k-1}\frac{1}{|\mathcal{M}_j|}\sum_{m\in\mathcal{M}_j}\|\nabla F_{m,j}\|^2 + 3S\sum_{j=k-S}^{k-1}\mathbb{E}\|\mathbf{e}_c(j)\|^2$$
$$\overset{(d)}{\leq} 3S^2\left(\delta^2+\alpha_1+\alpha_2\|\nabla F(\mathbf{x}(k))\|^2\right) + 3S\mathbb{E}\sum_{j=k-S}^{k-1}\|\mathbf{e}_c(j)\|^2,$$

where (a) is from $\|\sum_{i=1}^n \mathbf{a}_i\|^2 \leq \sum_{i=1}^n \|\mathbf{a}_i\|^2, \forall \mathbf{a}_i \in \mathbb{R}^d$, (b) is from Assumption 3, (c) is from Assumption 5, and (d) is from Assumption 4. Therefore, we have

$$\mathbb{E}\|\mathbf{e}_a(k)\|^2 \leq 6S^2\left(\delta^2+\alpha_1+\alpha_2\mathbb{E}\|\nabla F(\mathbf{x}(k))\|^2\right) + 6S\mathbb{E}\sum_{j=k-S}^{k-1}\|\mathbf{e}_c(j)\|^2 + 2\delta^2,$$

which completes the proof of Proposition 1.

Appendix B
Proof of Corollary 1

We expand the communication MSE in (23) and obtain

$$\mathbb{E}\|\mathbf{e}_c(k)\|^2 \overset{(a)}{=} \sum_{c=1}^{C}\mathbb{E}\Big[\sum_{m\in\mathcal{M}_k} 2\Delta_m(k)^2 + \zeta(k)^2\delta_n^2\Big], \quad (40)$$

where $\Delta_m(k) = \frac{\sqrt{v_m(k)}}{|\mathcal{M}_k|} - \zeta(k)|h_m(k)|b_m(k)$, and (a) is from $\mathbb{E}[\mathbf{n}(k)] = \mathbf{0}$ and $\mathbb{E}[r_m^{(k)}(c)r_m^{(k)}(c)^\dagger] = 2$. Note that (40) is a convex quadratic function with respect to $\zeta(k)$, and the optimal $\zeta(k)$ minimizing (40) is given by

$$\zeta(k) = \frac{\sum_{m\in\mathcal{M}_k}\frac{\sqrt{v_m(k)}}{|\mathcal{M}_k|}|h_m(k)|b_m(k)}{\sum_{m\in\mathcal{M}_k}|h_m(k)|^2 b_m(k)^2 + \delta_n^2/2}. \quad (41)$$

Substituting (41) into (40), we obtain

$$\mathbb{E}\|\mathbf{e}_c(k)\|^2 = 2\sum_{c=1}^{C}\mathbb{E}\left[\sum_{m\in\mathcal{M}_k}\frac{v_m(k)}{|\mathcal{M}_k|^2} - \varphi(k)\zeta(k)\right], \quad (42)$$

where $\varphi(k) = \sum_{m\in\mathcal{M}_k}\frac{\sqrt{v_m(k)}}{|\mathcal{M}_k|}|h_m(k)|b_m(k)$. In addition, (42) can be further bounded as

$$\mathbb{E}\|\mathbf{e}_c(k)\|^2 \overset{(a)}{=} \frac{2Cv_m(k)}{|\mathcal{M}_k|^2}\mathbb{E}\Big[|\mathcal{M}_k| - \varepsilon(k)\Big] \overset{(b)}{\leq} 2Cv_m(k)\mathbb{E}\Big[|\mathcal{M}_k| - \varepsilon(k)\Big] \overset{(c)}{\leq} \|\nabla f_m(\mathbf{x}(k-\tau_{m,k}))\|^2\mathbb{E}\Big[|\mathcal{M}_k| - \varepsilon(k)\Big] \overset{(d)}{\leq} \left(\delta^2+\alpha_1+\alpha_2\|\nabla F(\mathbf{x}(k))\|^2\right)\Big[|\mathcal{M}_k| - \varepsilon(k)\Big],$$

where $\varepsilon(k) = \frac{(\sum_{m\in\mathcal{M}_k}|h_m(k)|b_m(k))^2}{\sum_{m\in\mathcal{M}_k}|h_m(k)|^2 b_m(k)^2+\delta_n^2/2}$, step (a) is based on the assumption that the gradient variance of each device is equal, i.e., $v_1^{(k)} = v_2^{(k)} = \cdots = v_m^{(k)}$, step (b) is based on the fact that when $\mathbb{E}\|\mathbf{e}_c(k)\| \neq 0$, $|\mathcal{M}_k| \geq 1$, step (c) is from $\sum_{d=1}^D (g_m^{(k)}(d) - \bar{g}_m^{(k)})^2 = \sum_{d=1}^D((g_m^{(k)}(d))^2 - (\bar{g}_m^{(k)})^2) \leq \sum_{d=1}^D (g_m^{(k)}(d))^2 = \|\nabla f_m(\mathbf{x}(k-\tau_{m,k}))\|^2$, and step (d) is from Assumption 4 and 5. Thus, we obtain (28).

Appendix C
Proof of Proposition 2

As demonstrated in [28], Assumptions 1 and 2 can provide an upper bound of the loss function $F(\mathbf{x}(k+1))$ with respect to the recursion (21), as shown in the following lemma.

Lemma 1 [28, Lemma 2.1]: Under Assumptions 1 and 2, with learning rate $\lambda = 1/L$, we have

$$\mathbb{E}[F(\mathbf{x}(k+1))] \leq \mathbb{E}[F(\mathbf{x}(k)) - \frac{1}{2L}\|\nabla F(\mathbf{x}(k))\|^2 + \frac{1}{2L}\|\mathbf{e}(k)\|^2], \quad (43)$$

where the expectations are taken over the communication noise.

Then, we bound the overall MSE $\mathbb{E}\|\mathbf{e}(k)\|^2$ in (43) as

$$\mathbb{E}\|\mathbf{e}(k)\|^2 \overset{(a)}{\leq} 3\mathbb{E}[\|\mathbf{e}_c(k)\|^2] + 3\mathbb{E}[\|\mathbf{e}_d(k)\|^2] + 3\mathbb{E}[\|\mathbf{e}_a(k)\|^2] \overset{(b)}{\leq} \alpha_2 g(\mathbf{q}_k,\mathbf{a}_k,\mathbf{b}_k)\|\nabla F(\mathbf{x}(k))\|^2 + (\delta^2+\alpha_1)g(\mathbf{q}_k,\mathbf{a}_k,\mathbf{b}_k) + 6\delta^2, \quad (44)$$

where $\mathbf{a}_k \in \mathbb{R}^M$, $\mathbf{b}_k \in \mathbb{R}^M$ and $\mathbf{q}_k \in \mathbb{R}^3$ be the k-th column of $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{Q}$, respectively, $g(\mathbf{q}_k,\mathbf{a}_k,\mathbf{b}_k) = 18S^2 + 24(\frac{M-\sum_{m=1}^M a_m(k)}{M})^2 + 18S\sum_{j=k-S}^k(\sum_{m=1}^M a_m(j) - \frac{(\sum_{m=1}^M a_m(j)\sqrt{g_0}\|\mathbf{q}(j)-\mathbf{w}_m\|^{-1}b_m(j))^2}{\sum_{m=1}^M a_m(j)g_0\|\mathbf{q}(j)-\mathbf{w}_m\|^{-2}b_m(j)^2+\delta_n^2/2})$, (a) is from the triangle inequality, and (b) is based on (24), (25) and (28),

Substituting (44) into (43), and using $\|\nabla F(\mathbf{x}(k))\|^2 \geq 2\mu\left(F(\mathbf{x}(k)) - F(\mathbf{x}^\star)\right)$ (from Assumption 2), we have

$$F(\mathbf{x}(k+1)) - F(\mathbf{x}^\star) \leq \frac{1}{2L}\left[g(\mathbf{q}_k,\mathbf{a}_k,\mathbf{b}_k)(\delta^2+\alpha_1) + 6\delta^2\right] + \Big(1 - \frac{\mu}{L}\left[1 - \alpha_2 g(\mathbf{q}_k,\mathbf{a}_k,\mathbf{b}_k)\right]\Big) \times \left(F(\mathbf{x}(k)) - F(\mathbf{x}^\star)\right). \quad (45)$$

Recursively applying (45) for K times, we obtain (29).

Appendix D
Proof of Proposition 4

We introduce auxiliary variable $\mathbf{f} \in \mathbb{R}^K$ satisfying $f_k \geq g(\mathbf{a}_k), \forall k$, and $\mathbf{d} \in \mathbb{R}^K$ satisfying $d_k \leq \frac{(\sum_{m=1}^M a_m(k)|h_m(k)|b_m(k))^2}{\sum_{m=1}^M a_m(k)|h_m(k)|^2 b_m(k)^2+\delta_n^2/2}, \forall k$ and reformulate (37) as

$$\min_{\mathbf{A},\mathbf{p},\mathbf{d},\boldsymbol{\omega},\mathbf{y}} \quad \log\left(\sum_{k=1}^K e^{\mathbf{c}_k^{\mathrm{T}}\mathbf{y}}\right) + \frac{1}{\eta}\sum_{m=1}^M P(\mathbf{A},\bar{\mathbf{A}}) \quad (46a)$$
$$\text{s.t.} \quad f_k \geq g(\mathbf{a}_k), \quad \forall k, \quad (46b)$$
$$d_k \leq \frac{(\sum_{m=1}^M a_m(k)|h_m(k)|b_m(k))^2}{\sum_{m=1}^M a_m(k)|h_m(k)|^2 b_m(k)^2 + \delta_n^2/2}, \quad \forall k, \quad (46c)$$

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$$f_k \leq \min\left(\frac{2Lp_i - 6\delta^2}{\delta^2 + \alpha_1}, \frac{Lp_{K+k} + \mu - L}{\mu\alpha_2}\right), \quad \forall k, \tag{46d}$$

$$(31b), (31c), (32d). \tag{46e}$$

When (46) is minimized, at the optimal point (46b) and (46c) must be held with equality; hence problem (46) is equivalent to problem (37).

Problem (46) now includes non-convex constraints (32d) and (46c). Denote $\{\mathbf{A}^t, \mathbf{y}^t\}$ as the optimizer $\{\mathbf{A}, \mathbf{y}\}$ obtained at the previous inner iteration. Note that (32d) is a constraint where a linear function is less than a convex function. Since the first-order Taylor expansion serves as a global lower bound of any convex function, (32d) can be convexified with a surrogate function by taking the first-order Taylor expansion to its right-hand side (RHS). We thus obtain (38c).

Furthermore, the RHS of (46c) is convex because it is a convex quadratic-over-linear function under a composition with affine mapping [36, Section 3.2]. Similarly, (46c) can be convexified by taking the first-order Taylor expansion to the RHS, as shown in (38e).

We now complete the convexification of all non-convex constraints in problem (37). The transformed problem is given in (38). Note that all the non-convex functions and their surrogate functions (derived via first-order Taylor expansion) have identical values and gradients at local point $\{\mathbf{A}^t, \mathbf{y}^t\}$. Moreover, the transformed problem satisfies the Slater's condition. According to [40, Section III-B], the solution of (38) terminates at a KKT point of (37) and qualifies as a local minimum if it resides in the interior of (38)'s feasible set.

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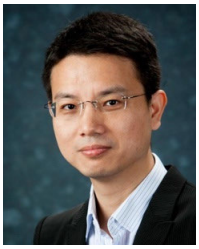
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Zhiyuan Zhai received the B.S. degree in communication engineering from the School of Information and Communication Engineering, University of Electronic Science and Technology of China, in 2022. He is currently pursuing the Ph.D. degree with the Department of Communication Science and Engineering, Fudan University, Shanghai, China. His research interests include machine learning, signal processing, and mobile edge computing.



Xiaojun Yuan (Senior Member, IEEE) received the Ph.D. degree in electrical engineering from the City University of Hong Kong in 2009. From 2009 to 2011, he was a Research Fellow with the Department of Electronic Engineering, City University of Hong Kong. He was also a Visiting Scholar with the Department of Electrical Engineering, University of Hawaii at Manoa, in Spring and Summer 2009 and in 2010. From 2011 to 2014, he was a Research Assistant Professor with the Institute of Network Coding, The Chinese University of Hong Kong. From 2014 to 2017, he was an Assistant Professor with the School of Information Science and Technology, ShanghaiTech University. He is currently a State-Specially-Recruited Professor with the National Key Laboratory on Wireless Communications, University of Electronic Science and Technology of China. His research interests cover a broad range of signal processing, machine learning, and wireless communications, including but not limited to intelligent communications, structured signal reconstruction, Bayesian approximate inference, and distributed learning. He has published over 200 peer-reviewed research papers in the leading international journals and conferences in the related areas. He has served on a number of technical programs for international conferences. He is an Editor of IEEE TRANSACTIONS ON WIRELESS COMMUNICATIONS. He was a co-recipient of the Best Paper Award of IEEE International Conference on Communications (ICC) 2014 and also a co-recipient of the Best Journal Paper Award of IEEE Technical Committee on Green Communications and Computing (TCGCC) 2017.



Xin Wang (Fellow, IEEE) received the B.Sc. and M.Sc. degrees in electrical engineering from Fudan University, Shanghai, China, in 1997 and 2000, respectively, and the Ph.D. degree in electrical engineering from Auburn University, Auburn, AL, USA, in 2004.

From September 2004 to August 2006, he was a Post-Doctoral Research Associate with the Department of Electrical and Computer Engineering, University of Minnesota, Minneapolis. In August 2006, he joined the Department of Electrical Engineering, Florida Atlantic University, Boca Raton, FL, USA, as an Assistant Professor, and then was promoted to a tenured Associate Professor in 2010. He is currently a Distinguished Professor and the Chair of the Department of Communication Science and Engineering, Fudan University. His research interests include stochastic network optimization, energy-efficient communications, cross-layer design, and signal processing for communications. He served as a Senior Area Editor for IEEE TRANSACTIONS ON SIGNAL PROCESSING, as an Associate Editor for IEEE TRANSACTIONS ON SIGNAL PROCESSING, as an Editor for IEEE TRANSACTIONS ON WIRELESS COMMUNICATIONS, as an Editor for IEEE TRANSACTIONS ON VEHICULAR TECHNOLOGY, and as an Associate Editor for the IEEE SIGNAL PROCESSING LETTERS. He is a member of the Signal Processing for Communications and Networking Technical Committee of IEEE Signal Processing Society and a Distinguished Speaker of the IEEE Vehicular Technology Society.



Huiyuan Yang (Graduate Student Member, IEEE) received the B.S. degree in communication engineering from the University of Electronic Science and Technology of China, Chengdu, China, in 2019, where he is currently pursuing the Ph.D. degree. His current research interests include information theory, machine learning, and wireless communication.

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